

AI Education at Scale

What can we learn from data science?

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AI at Sciencepark • Amsterdam

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What we learned from Data 8

- Start with **doing**, not prerequisites
- Real questions + real datasets
- Short computational explorations
- Introduce programming, statistics and concepts as needed
- Use the **real tools**
- Reach students outside computer science

We didn't make students learn all the machinery before we let them do data science.



Lecture 1

Introduction

Data 8 · Introduction, Spring 2018

The notebook became the laboratory

Browser → Notebook → Data → Experiment

- Jupyter + Python + real data
- Change code → immediately see what happens
- Same environment for instructors and students
- Reproducible across a class
- Works for 20 students — or thousands
- Open-source tools + open curriculum + shared infrastructure

What if we did the same thing with AI?

Put a model in the notebook.

- Prompt it
- Change parameters
- Examine tokens and probabilities
- Create embeddings
- Add retrieval
- Compare models
- Build evaluations
- Investigate failures

Small models are enough to learn a lot

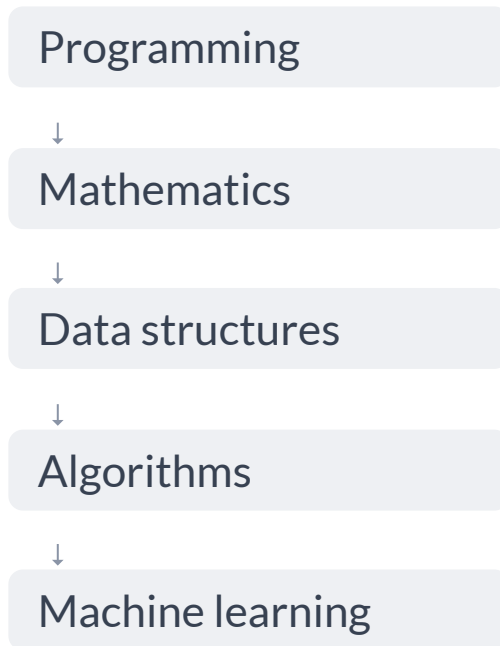
You don't need a frontier model for every experiment.

- 1B–7B parameter models
- Downloadable
- Run locally or on shared infrastructure
- CPU for some experiments; GPUs when needed
- Instructor-selected and tested
- Reproducible across a class
- **Actual AI models — not educational simulations**

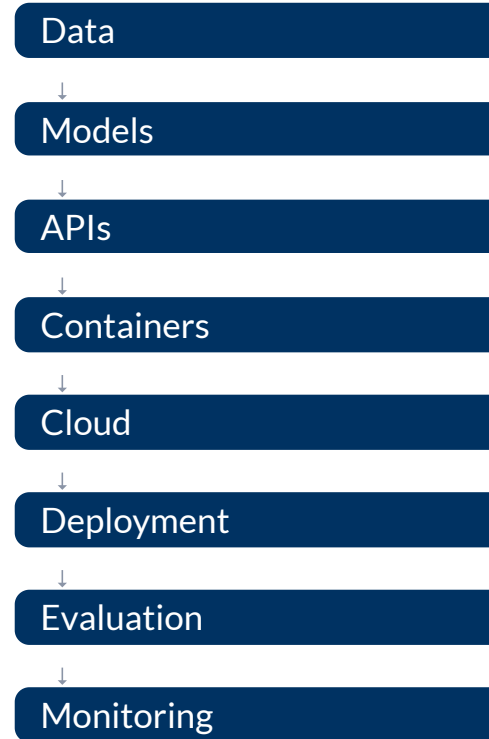
But I think something else is changing

What counts as technical literacy?

Traditional computing / ML



Modern AI workflow



Are we moving from teaching **algorithms** to teaching **systems**?

Students need to see the whole stack

A student can understand **gradient descent** but have no idea how an AI application actually works.

A student can now build an **AI application** without understanding gradient descent.

Both are educational problems.

- Data engineering
- APIs
- Containers and reproducible environments
- Cloud / GPUs / inference
- MLOps / LLMOps
- Evaluation and monitoring

Teach enough engineering that the system stops looking like magic.

Who owns the laboratory?

Commercial AI

Notebook



API



Proprietary model



Commercial cloud

Student learns to **consume** AI

Open AI

Notebook



Open-source tooling



Open model



Shared / public compute

Student can **inspect, modify, evaluate and build**

Small + Open + Public

A European opportunity?

Europe doesn't necessarily have to win the race for the **biggest model**.

Perhaps it can compete differently.

- Smaller and more efficient models
- Open weights
- Open-source software
- Shared public compute
- University and research infrastructure
- Open educational materials
- People who understand the entire stack

Can Europe build a lighter-weight, open AI ecosystem rather than simply reproducing the hyperscalers?

What should the university build?

Students across the disciplines

Open curriculum

Interactive notebooks

Open-source tooling

Open models

Public / shared compute

Simple for students and instructors: open a notebook, choose a model, run an experiment.

Substantial infrastructure underneath, built and shared by universities.

Three ideas

1. Interactive pedagogy

We know how to make technical ideas accessible through **exploration**.

2. Systems literacy

Students need to understand **workflows**, not only algorithms.

3. Open infrastructure

Universities should help students become **builders**, not simply customers.